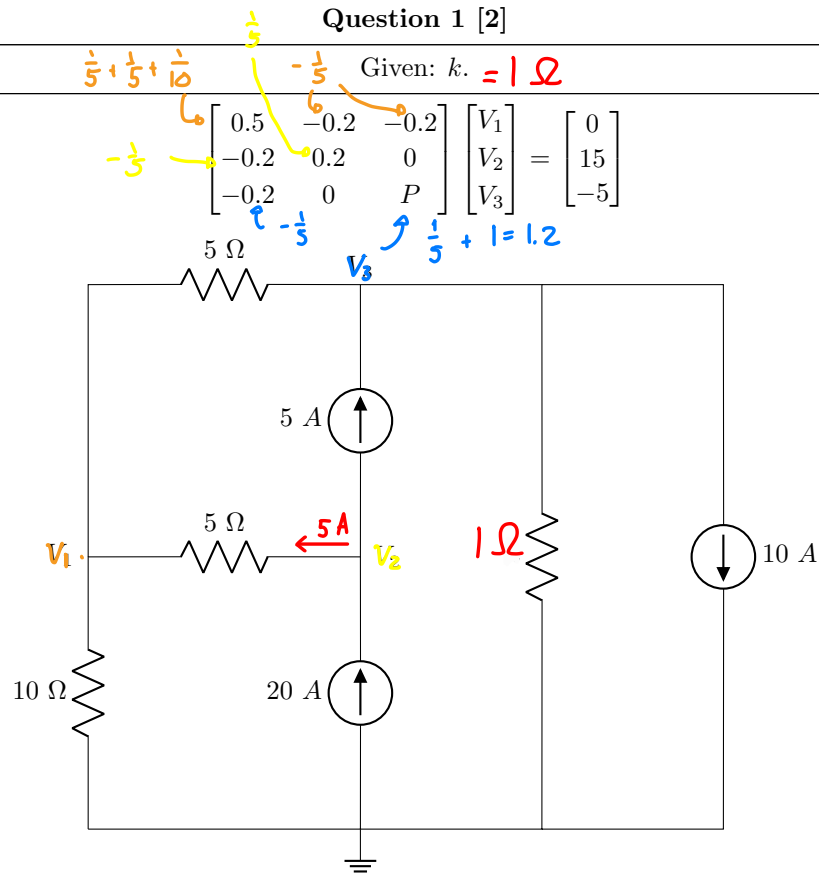


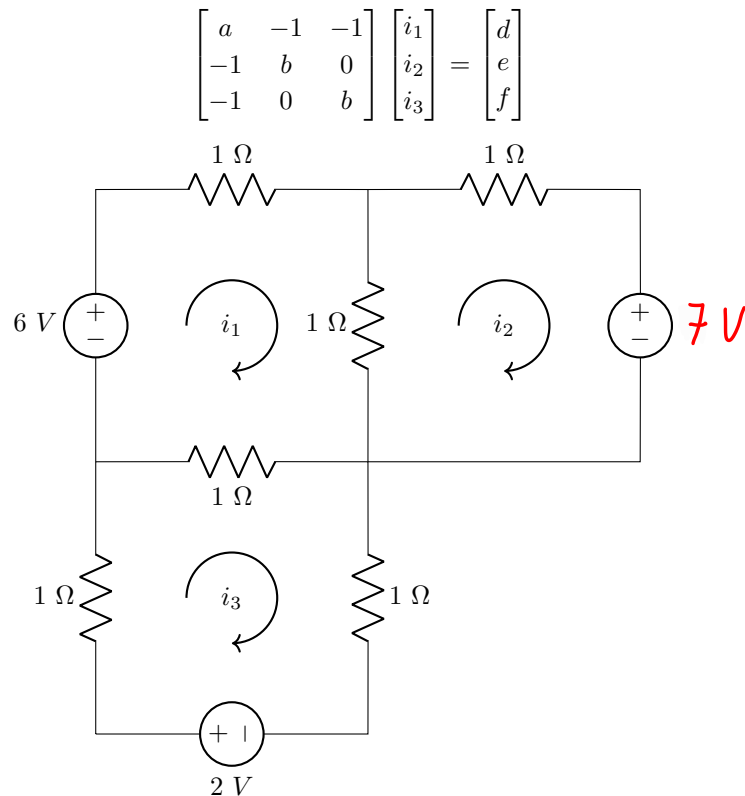
Section 1

Question 1 [2]



01 Determine P in the matrix via inspection. [2]

$$P = \frac{1}{5} = 1.2$$

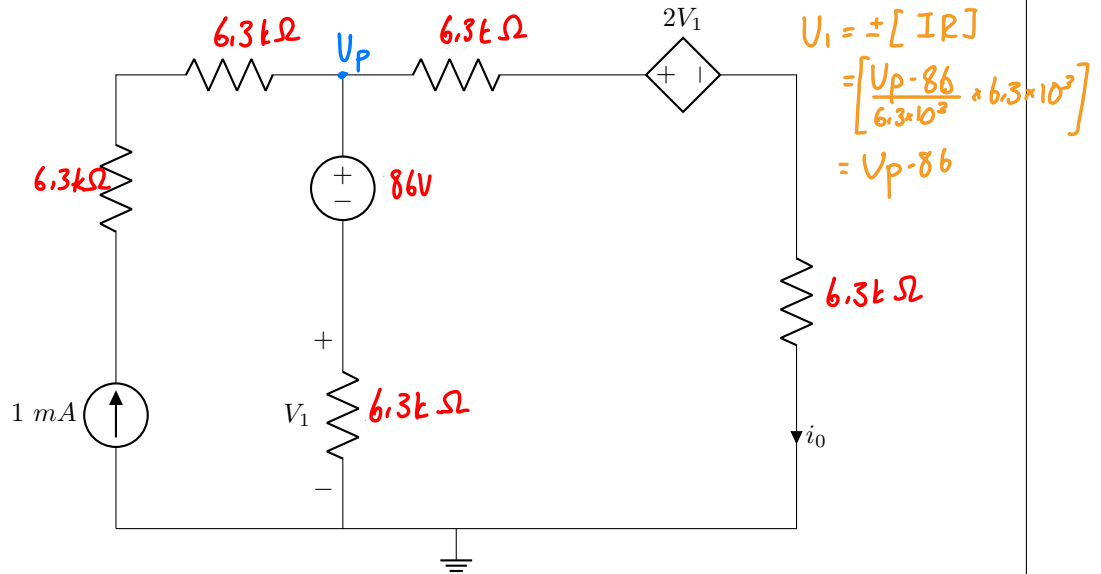
Question 2 [2]Given: $V_1 = 7$ **02** Find e by inspection. [2]

$$\begin{bmatrix} 3 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & 0 & 3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -7 \\ 2 \end{bmatrix}$$

$$\underline{e = -7}$$

voltages: if enters through - terminal
it's positive

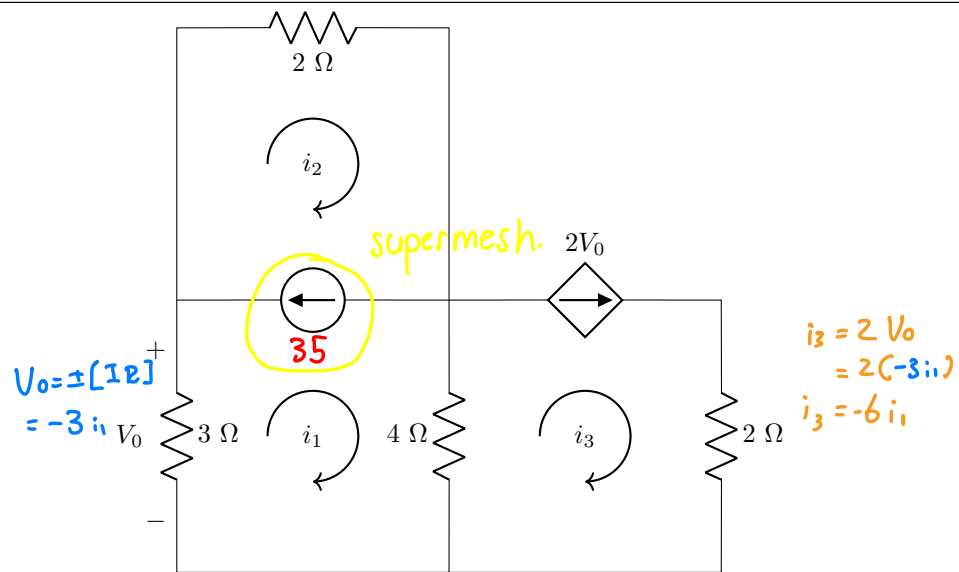
Question 3 [2]

Given: $R_1 = R_2 = R_3 = R_4 = R_5 = 6.3 \text{ k}\Omega$, $V_1 = 86 \text{ V}$ 03 Find i_0 using either nodal or mesh analysis. [2]

$$\begin{aligned} @V_p: \frac{V_p - 86}{6.3 \times 10^3} + \frac{V_p - 2V_1}{12.6 \times 10^3} - 1 \times 10^{-3} &= 0 \\ \frac{V_p - 86}{6300} + \frac{V_p - 2V_p + 172}{12600} &= 1 \times 10^{-3} \\ 2V_p - 172 - V_p + 172 &= 12.6 \\ V_p &= 12.6 \end{aligned}$$

$$\begin{aligned} i_0 &= \frac{V_p - 2V_1}{6300} \\ &= \frac{V_p - 2(V_p - 86)}{6300} \\ &= \frac{12.6 - 2(12.6 - 86)}{6300} \\ i_0 &= 25.30 \text{ mA} \end{aligned}$$

Question 4 [2]

Given: $k = 35 A$ 04 Determine i_2 . [2]

$$\begin{array}{l}
 \uparrow I_2 \\
 \leftarrow 35 \\
 \uparrow I_1
 \end{array}
 \quad
 \begin{array}{l}
 I_{in} = I_{out} \\
 35 + I_1 = I_2 \\
 I_1 - I_2 = -35 \quad I_1 = I_2 - 35
 \end{array}$$

loop 1 + loop 2 (mesh analysis)

$$3I_1 + 4(I_1 - I_3) + 2I_2 = 0$$

$$7I_1 + 2I_2 - 4I_3 = 0$$

$$7I_1 + 2I_2 + 24I_1 = 0$$

$$31I_1 + 2I_2 = 0$$

$$31(I_2 - 35) + 2I_2 = 0$$

$$31I_2 - 1085 + 2I_2 = 0$$

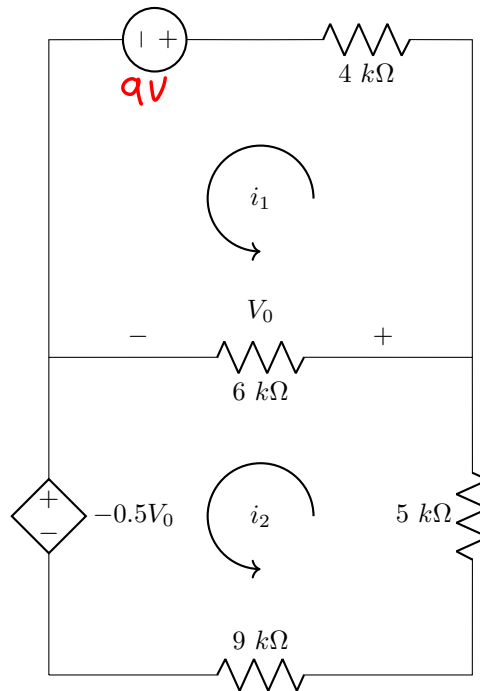
$$33I_2 = 1085$$

$$\underline{I_2 = 32.884}$$

Question 5 [2]Given: $V_x = 9V$

Use mesh analysis to set up an equation in the following form:

$$10i_1 - 6i_2 = c$$

**05** Determine c [2]

$$@ i_1: 9 + 4000i_1 + 6000i_1 - 6000i_2 = 0$$

$$10k i_1 - 6k i_2 = -9$$

$$10i_1 - 6i_2 = -9 \times 10^{-3}$$

$$c = -9 \text{ mV}$$

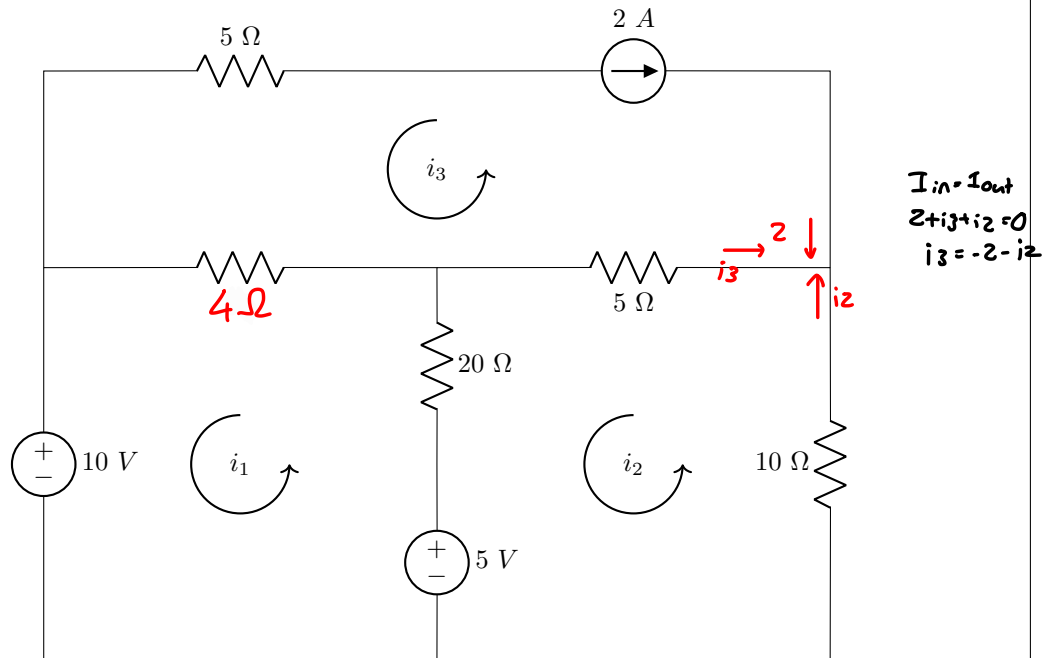
Question 6 [2]

Given: $R_v = 4\ \Omega$

Use mesh analysis to set up two equations in the following form:

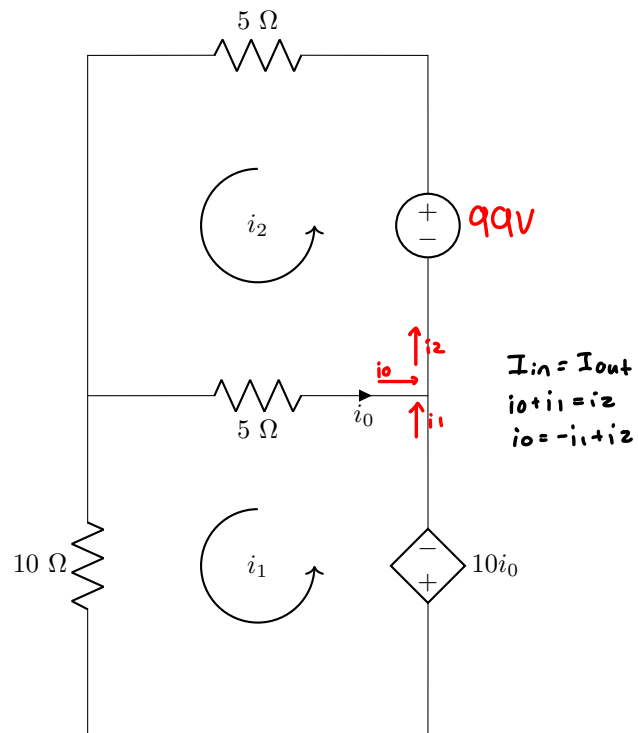
$$a\ i_1 - 20\ i_2 = b$$

$$-20\ i_1 + 35\ i_2 = -15$$

06 Determine b [2]

@ i_1 : $10 - 5 + 20i_1 - 20i_2 + 4i_1 - 4i_3 = 0$
 $24i_1 - 20i_2 - 4i_3 = -5$
 $i_3 = -2$
 $24i_1 - 20i_2 + 8 = -5$
 $24i_1 - 20i_2 = -13$
 $b = -13$

Question 7 [2]

Given: $V = 99V$ 07 Using mesh analysis, find i_1 [2]

@ i_1 :

$$10i_1 + 5i_1 - 5i_2 + 10i_0 = 0$$

$$15i_1 - 5i_2 - 10i_1 + 10i_2 = 0$$

$$5i_1 + 5i_2 = 0$$

$$i_1 = -i_2$$

@ i_2 :

$$-99 + 5i_2 + 5i_2 - 5i_1 = 0$$

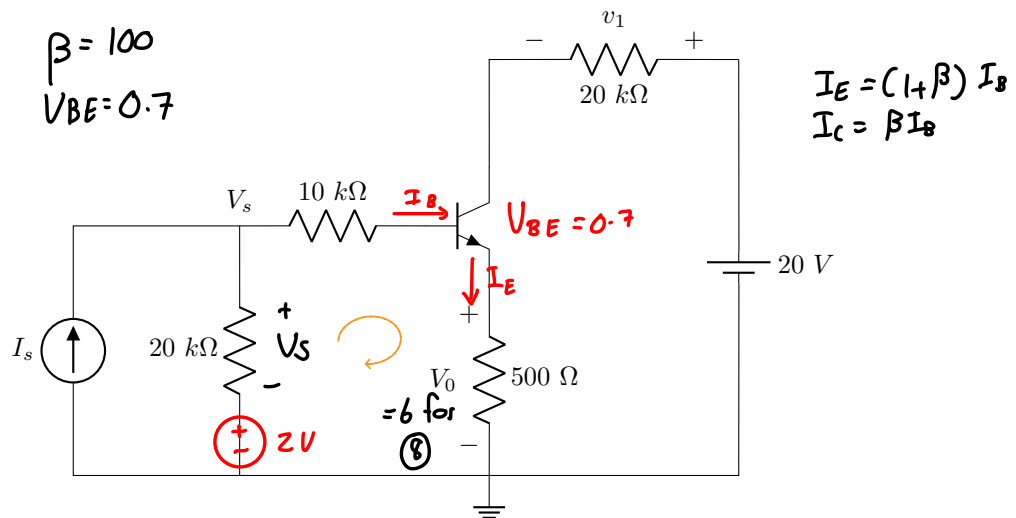
$$10i_2 - 5i_1 = 99$$

$$10i_2 + 5i_2 = 99$$

$$i_2 = 6.6A$$

$$\therefore \underline{i_1 = -6.6A}$$

Questions 8 to 9 [4]

Given: β , $V_{BE} = 100, 0.7 V$ Given: $V_0 = 6V$ 08 Determine V_s [2]Given: $I_s = 0.1 mA$ 09 Determine V_0 [2]

⑧

$$V_0 = I_E(500)$$

$$6 = I_E(500)$$

$$I_E = 12 mA$$

$$I_B = \frac{I_E}{100 + 1}$$

$$I_B = 118.81 \mu A$$

$$KVL: -V_s + 10 \times 10^3 \times 118.81 \times 10^{-6} + 0.7 + 6 = 0$$

$$\underline{V_s = 7.889V}$$

⑨ Source transform

$$0.1 \times 10^{-3} \times 20 \times 10^3 = 2V$$

$$KVL: -2 + 20 \times 10^3 I_B + 10 \times 10^3 I_B + 0.7 + 500 I_E = 6$$

$$30 \times 10^3 I_B + 500 I_E = 1.3$$

$$I_B = \frac{I_E}{101}$$

$$30 \times 10^3 \frac{I_E}{101} + 500 I_E = 1.3$$

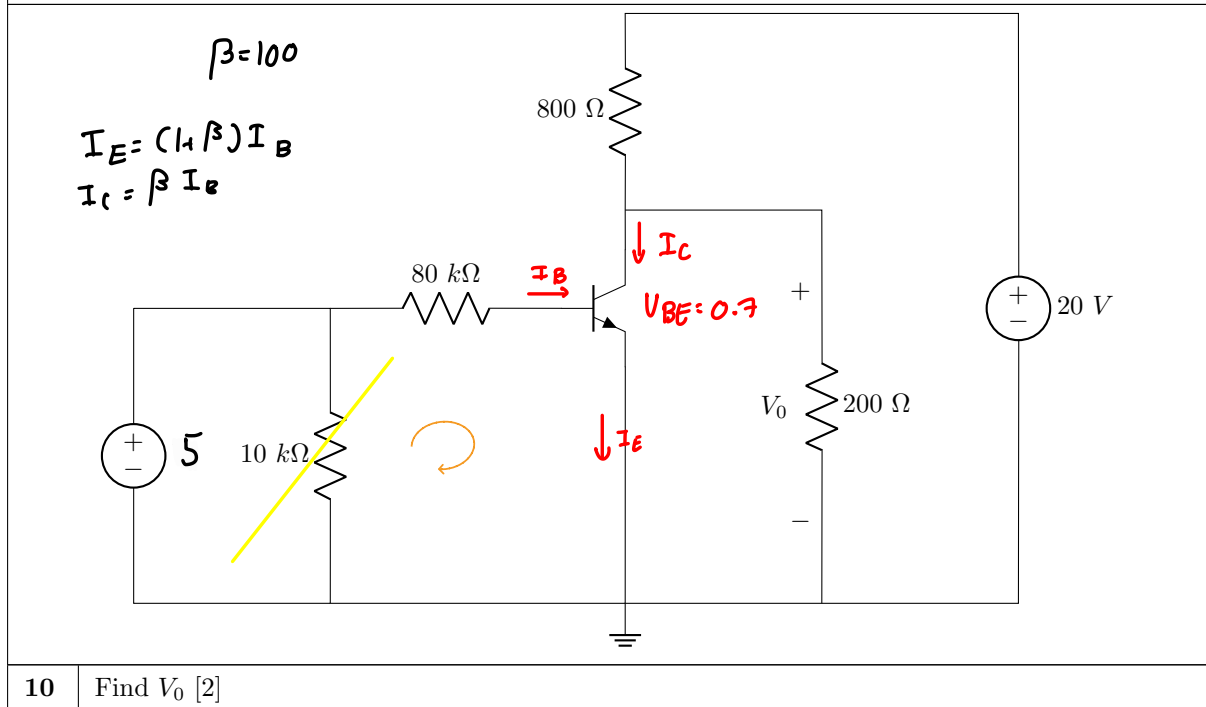
$$I_E \left(\frac{30 \times 10^3}{101} + 500 \right) = 1.3$$

$$I_E = 1631 \mu A$$

$$V_0 = 500 \times I_E$$

$$= 815.53 mV$$

Question 10 [2]

Given: V_s , β , $V_{BE} = 5V, = 100, 0.7V$ 

KVL: $-5 + 80 \times 10^3 I_B + 0.7 = 0$

$$80 \times 10^3 I_B = 4.3$$

$$I_B = 53.75 \mu A$$

$$I_C = 100 \times 53.75 \times 10^{-3}$$

$$I_C = 5.375 \text{ mA}$$

nodal:

$$\frac{V_0}{200} + \frac{V_0 - 20}{800} + 5.375 \times 10^{-3} = 0$$

$$4V_0 + V_0 - 20 = -4.3$$

$$5V_0 = 15.7$$

$$\underline{V_0 = 3.14 \text{ V}}$$

Total Section 1: [20]

Grand Total : [20]